

ECE 383 – Embedded Systems II

Final Project Format Guidelines

1 Proposal

Your final project proposal outlines the requirements your solution must meet, and the behavior of your solution. The proposal will start out by defining the name of the project and your name (with signatures) on the project. The following is the suggested format.

The Bicycle Computer A final project proposal for ECE 383 by C3C Cadet Jones _____ March 10, 2020
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The project proposal will then have the following three sections in the following order.

1.1 Objective Statement

The objective statement defines the purpose of the system you are designing and the need the project supposes to fulfill.

1.2 Requirements

Define the requirements that you will need to be met in order to achieve minimum functionality. Likewise define B and A-level functionality.

In addition, you must incorporate an external device you haven't previously used in this course to achieve B-level functionality. This could be the NES controller or another device of your choosing but the device must be more than a simple switch or button. Other options include other game peripherals (e.g. joysticks, other controllers, racing wheels), IR remotes, MIDI keyboards, sensors (e.g. temperature, light, sound, distance, cameras). Your instructor must approve the device.

Minimum functionality is 20 points, and B or A functionality are additional 10 points each, totaling to 40 points overall for functionality

You can earn 10 to 25 additional points based on the "difficulty factor" of your project. Since the NES controller is well documented, it is worth less points, while a new device or the IR controller is worth more points.. The other 10 points will be for extra features you implement above the minimum functionality. These 40 points will be weighted by a "difficulty factor". Those students with a more challenging project can receive more points.

1.3 Level-0 Description & Top Level Design

A level-0 function table defines the overall input, output, and behavior of the system that you are building. You need to carefully consider what interactions a user will have with your system. This will give you clear goals for your project.

- **Graphical** - Draw your system with a box for each major subsystem (For example, FPGA, Input Sensors, Output Displays) and with the top level signal inputs and outputs between the subsystems. Have a specific line for each input/output signal, defining what the signal is (i.e., temperature, velocity, etc), what the specific interface link is (i.e., USB, Bluetooth, etc), and what port it connects to on each device (i.e., UART port, GPIO pin, etc). For the FGPA, draw internal blocks to delineate if the system is just Custom Hardware (FSM and Datapath), just Microblaze, or a combination of Microblaze and custom hardware.

As you think about what you are proposing for your requirements and this design, you might want to consider some of these “hint” files on the class website (under the datasheet tab)

Final Project Help

- [IR controller hints](#)
- [NES controller hints](#)
- [PS2 Mouse interface hints](#)
- [Example Grid Memory\(zip\)](#)
- [Example Graphics Memory, 2bits per pixel\(zip\)](#)
- [Reading UART keyboard without being locked out](#)
- [Analog to Digital Converter hints](#)
- [Random Number Hints](#)
- [How to 3D print a case for your FPGA board](#)

You might also want to look over the list of past projects (some with videos)

[Final Project Ideas](#)

2 Plan

The project plan defines how you are going to go about implementing the design set forth in your proposal. The plan should then go on to include the following sections.

2.1 Proposal

Include the body of the updated proposal document - include any edits made to the original proposal based on feedback from your instructor, included updated requirements and functionality.

2.2 Detailed Architecture and Sub-System Design

You need to provide the detailed design of your system. A detailed design should be split into level-1 subsystems, such as datapath and control.

2.2.1 Level-1 Design

A level-1 design breaks the level-0 design into further subsystems and modules; each module generally corresponds to some physical chip, sensor, or actuator. A system boundary should be drawn showing where the system interacts with the user (signals which cross this boundary are the I/O from the level-0 function table). Example subsystems within your FPGA include Datapath, Control (FSM), Microblaze, and Key I/O ports (like UART, Bluetooth, etc). Describe the internal organization of the chip using a datapath and control similar to those presented in Lab 1, 2, and 4. Your datapath should be composed of several basic building block (BBBs), like in Lab 1, 2, and 4. Make sure that the top level-0 signals match those in the Level-1 diagram.

The Level-1 Design should include a block diagram, showing all I/O signals and interfaces between the subsystems. Subsystems requiring software should have either a flowchart or FSM design.

2.3 Calculations/Analysis/Drawings

You may need to perform some calculations, equations, or analysis in order for your project to work or to derived the key specifications for your system. Any unusual data structures which have bit fields which meet some requirements should be explained in this section, e.g. fixed point. Any math functions implemented in look-up-tables should be defined.

If your system is creating images on the HDMI monitor (like scopeface), you should include a drawing of the example scopeface display with the objects and game area drawn to scale (row/col pixel scale)

3 Final Demonstration and Test Results

This section should document the test and demonstration results the overall system was able to achieve.

4 Write-up

The final write-up should have sections 2 to 3 above. If you have modified your milestone goals, functionality goals, Level-0 or Level-1 Designs since your proposal/plan, then these should be updated in your final report.

The write-up should be posted in your GitHub repo in either markdown, word, or pdf format.

5 GitHub Repo

You should provide the faculty member with a BitBucket repository containing four directories. These four directories can be included in your current ECE 383 repository in a folder called FINAL PROJECT. SOURCE should contain all the project files (include all intermediate files generated by the compiler), PRESENTATION should contain your final power point presentation, DEMO should contain a mpeg, mov, avi, etc. of your project in action along with any documented tests (or a link to a video hosting site like youtube), and REPORT should contain your report.